

UNIT 4: Structured populations

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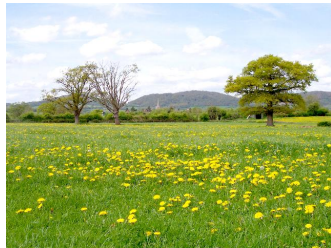
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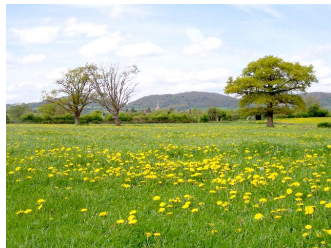
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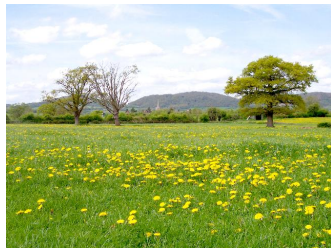
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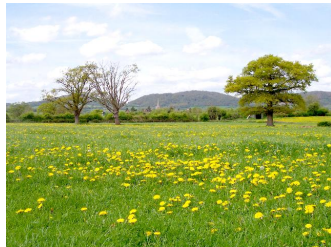
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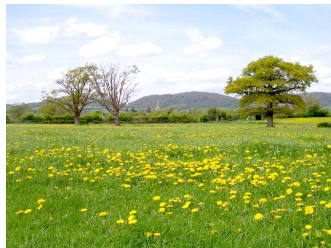
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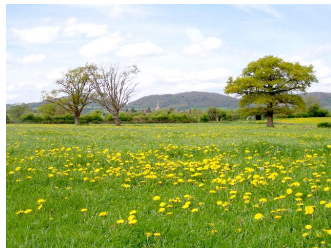
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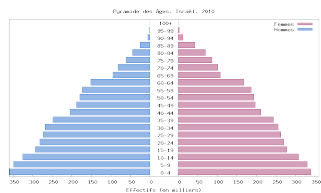
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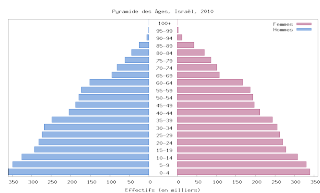
Modeling approach

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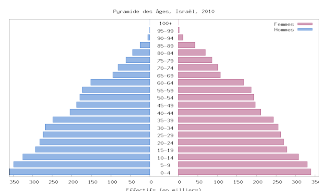
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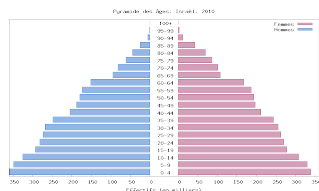
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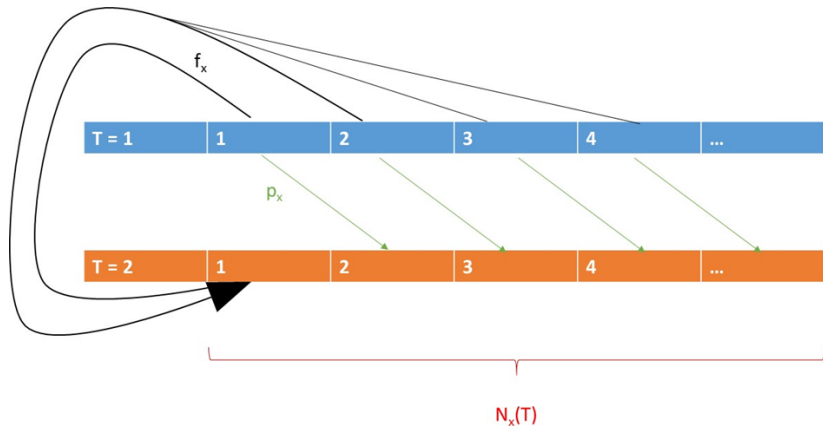
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x	f_x	p_x	l_x	$l_x f_x$
1				
2				
R				

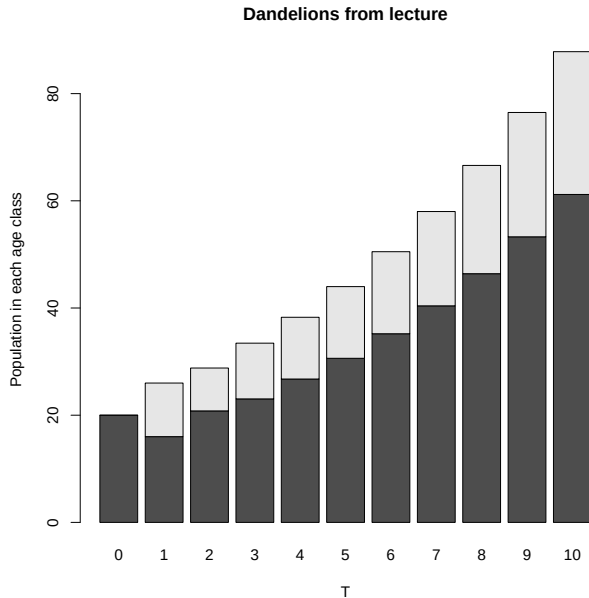
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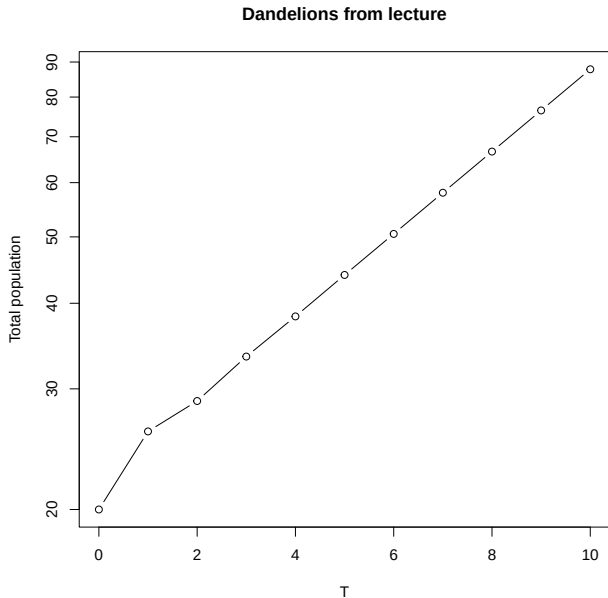
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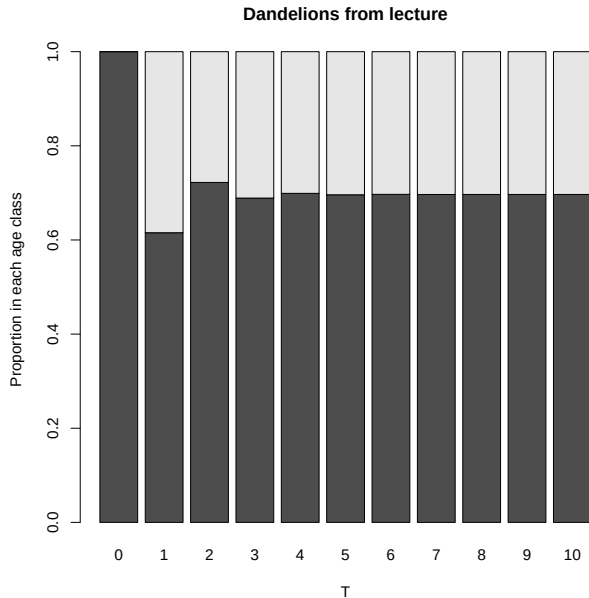
Dandelion dynamics



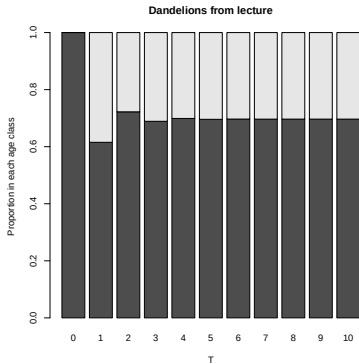
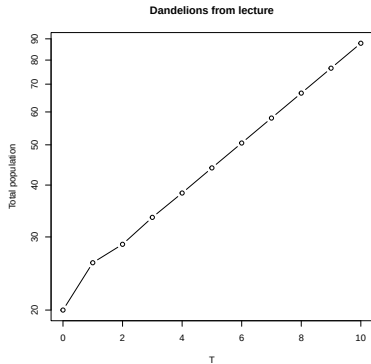
Dandelion population dynamics



Dandelion age dynamics



Dandelion dynamics



Squirrel example



Gray squirrel population example

x	f_x	p_x	ℓ_x	$\ell_x f_x$
1	0	0.25		
2	1.28	0.46		
3	2.28	0.77		
4	2.28	0.65		
5	2.28	0.67		
6	2.28	0.64		
7	2.28	0.88		
8	2.28	0.0		
R				

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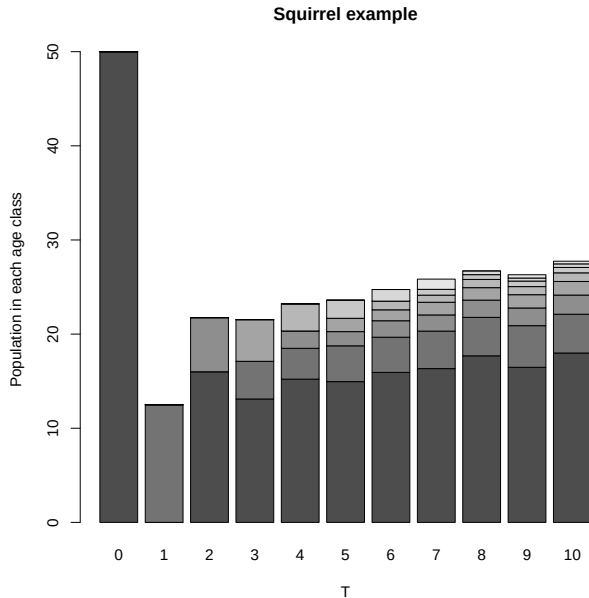
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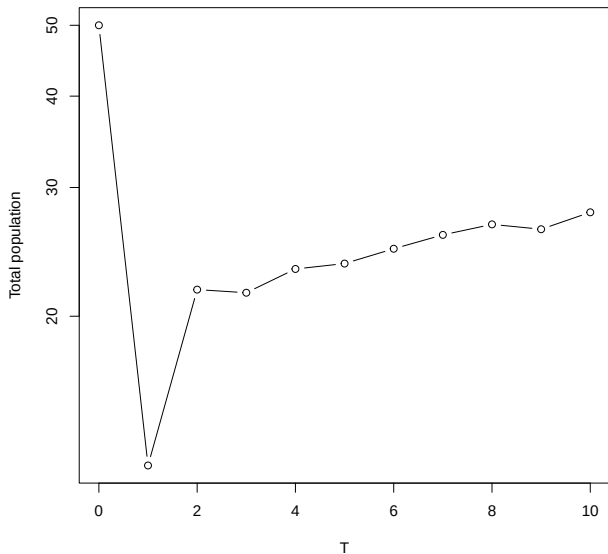
x	f_x	p_x	ℓ_x	$\ell_x f_x$
1	0	0.25	1.000	0.000
2	1.28	0.46	0.250	0.320
3	2.28	0.77	0.115	0.262
4	2.28	0.65	0.089	0.202
5	2.28	0.67	0.058	0.131
6	2.28	0.64	0.039	0.088
7	2.28	0.88	0.025	0.056
8	2.28	0.0	0.022	0.050
R				1.109

Gray squirrel dynamics

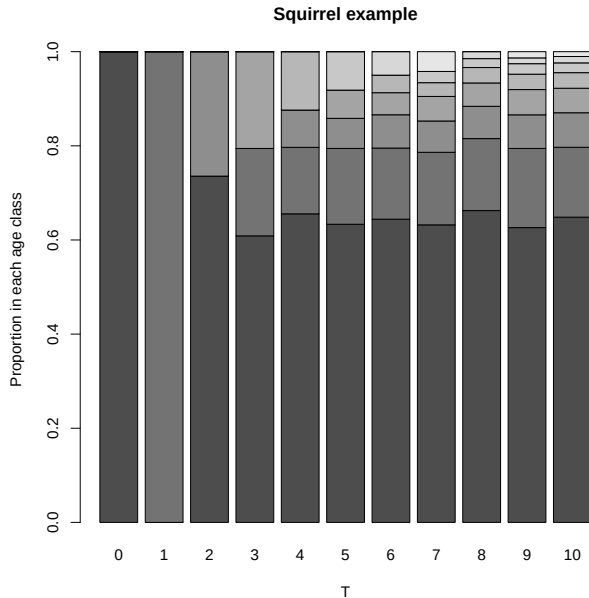


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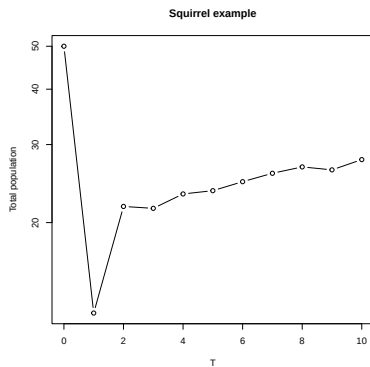
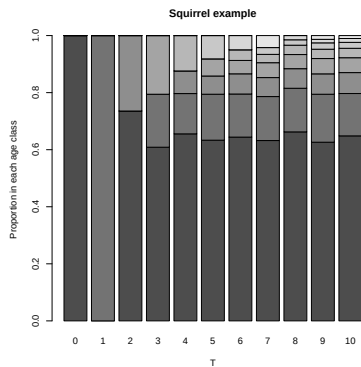
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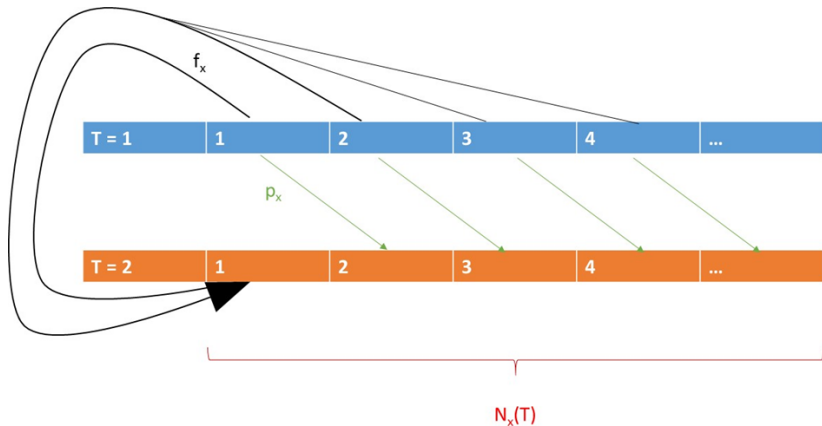
Gray squirrel age dynamics



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The structured model



Salmon example

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QSLIDE Salmon example

x	f_x	p_x	ℓ_x	$\ell_x f_x$
1	0	0.1		
2	0	0.5		
3	0	0.6		
4	50	0		
R				

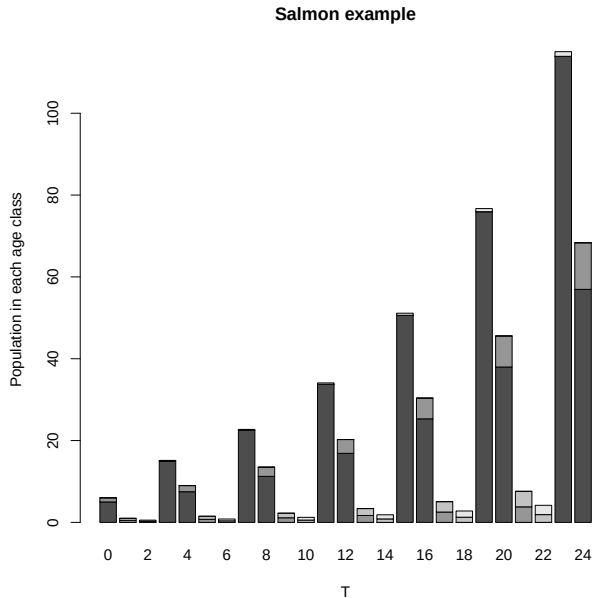
ASLIDE Salmon example

x	f_x	p_x	ℓ_x	$\ell_x f_x$
1	0	0.1	1.000	0.000
2	0	0.5	0.100	0.000
3	0	0.6	0.050	0.000
4	50	0	0.030	1.500
R				1.500

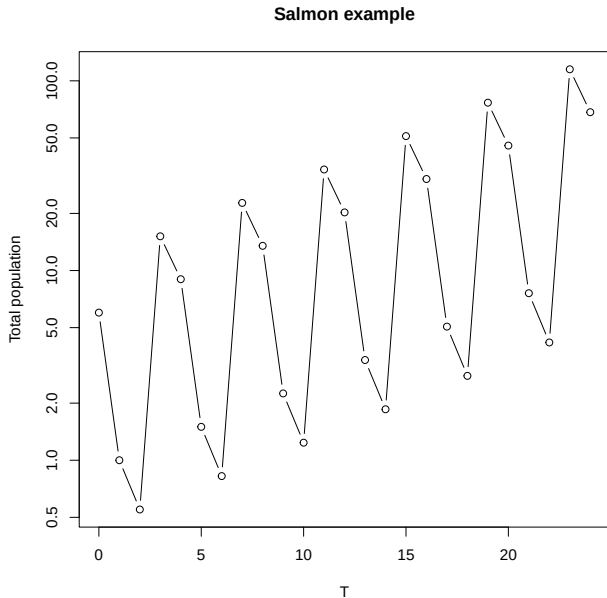
PSLIDE Salmon example



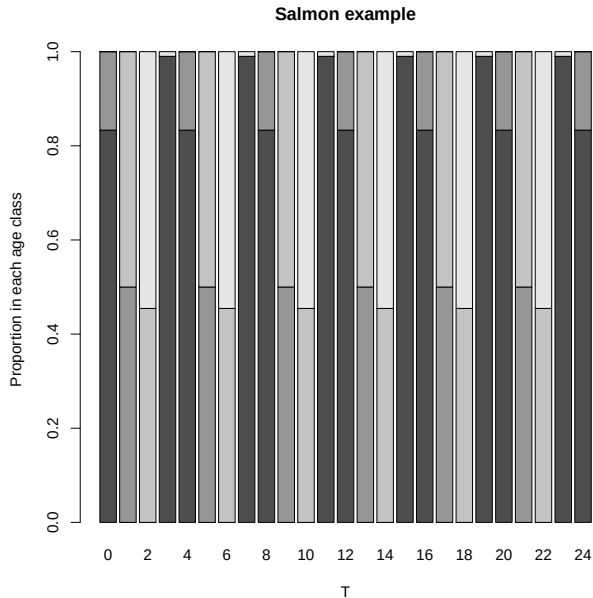
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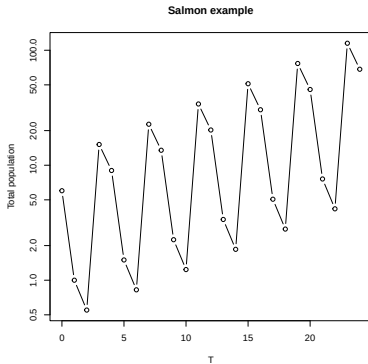
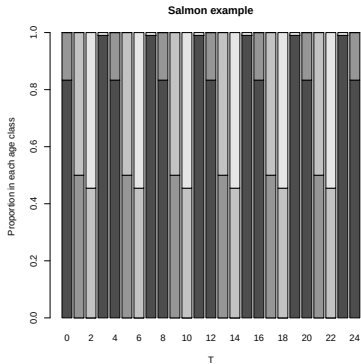
PSLIDE Salmon population dynamics



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Dandelion life table

x	f_x	p_x	l_x	$l_x f_x$
1	0.8	0.5	1.000	0.800
2	0.8	0	0.500	0.400
R				1.200

Counting after reproduction

x	f_x	p_x	ℓ_x	$\ell_x f_x$
1	0.8	0.01	1.000	0.800
2	40	0	0.010	0.400
R				1.200

There are two different approaches to the third age class: if we assume that we count the two-year old adults ($x = 3$), we can write $p_2 = 0.5$; $f_3 = 0$, and get the same answer (with one extra row that has zero contribution).

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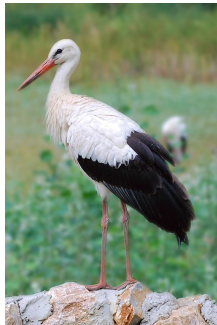
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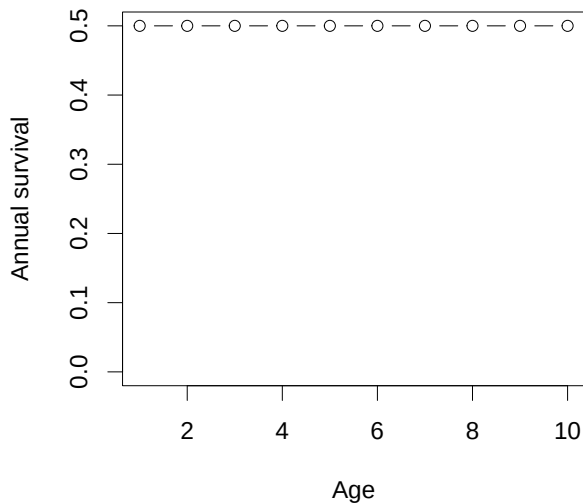
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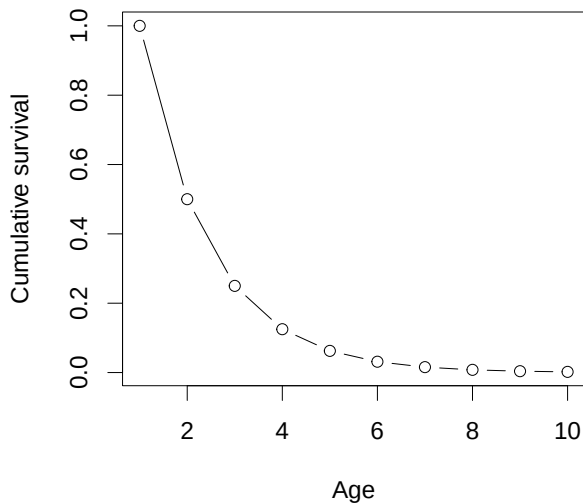
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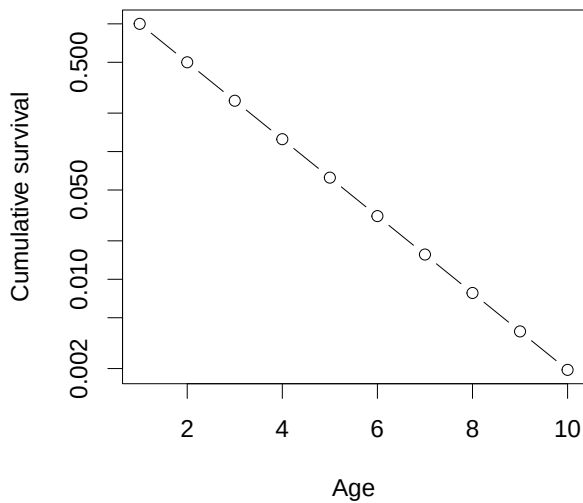
Constant survivorship



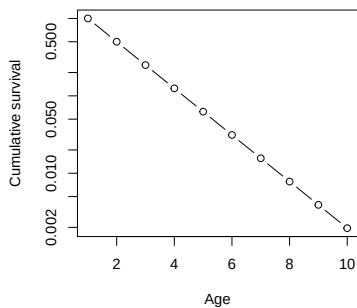
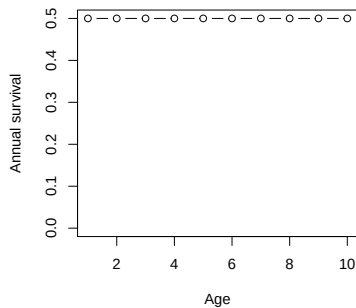
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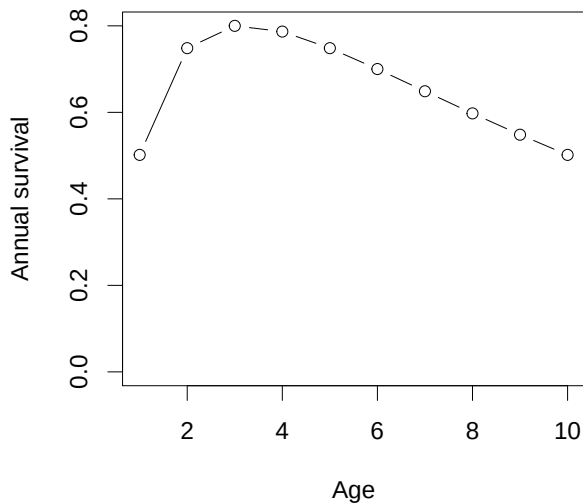
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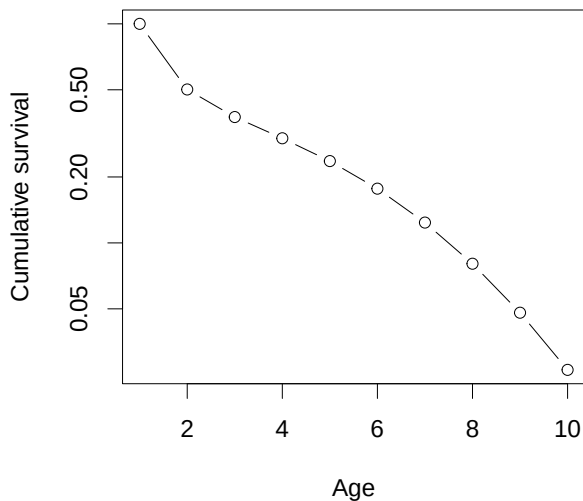
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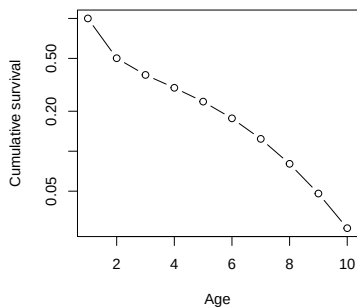
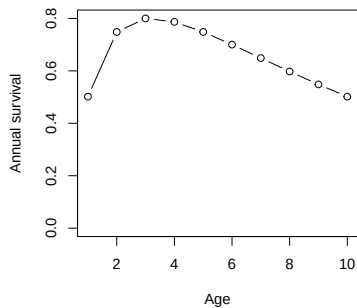
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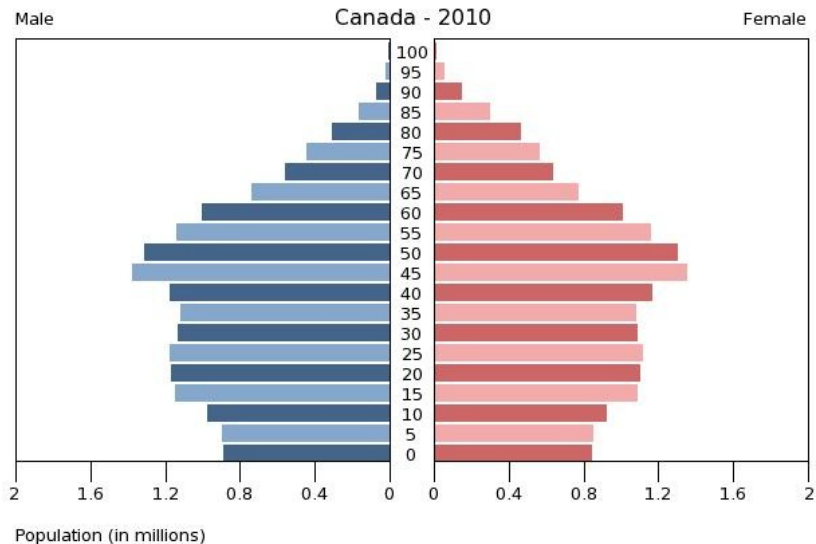
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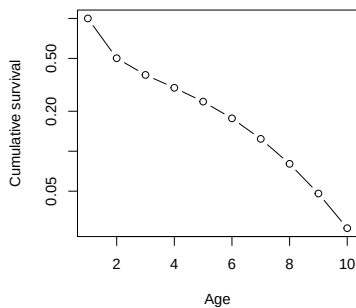
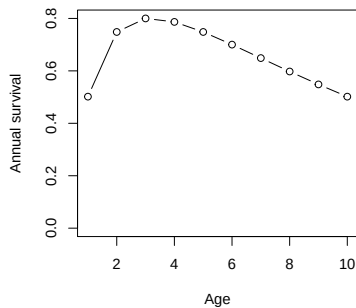
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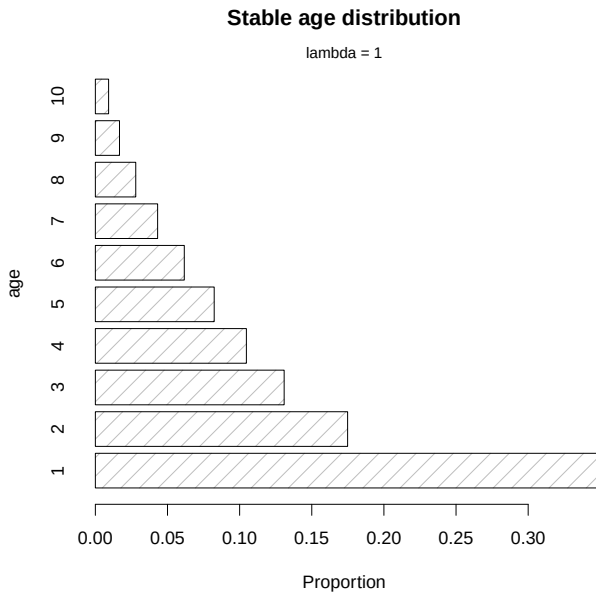
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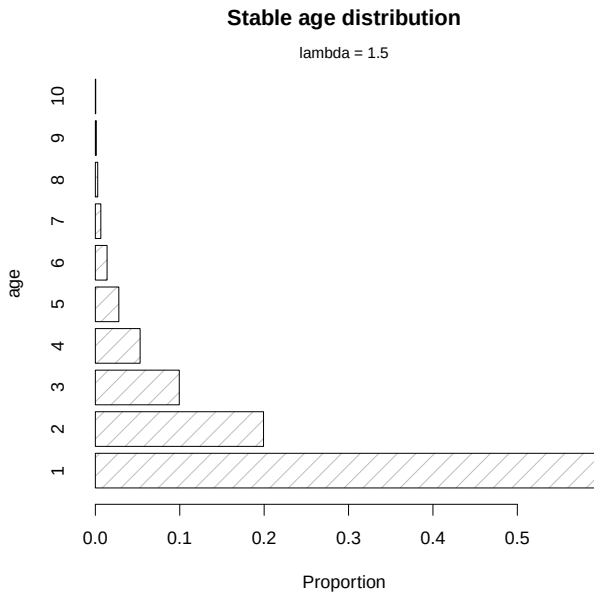
REPSLIDE Changing survivorship



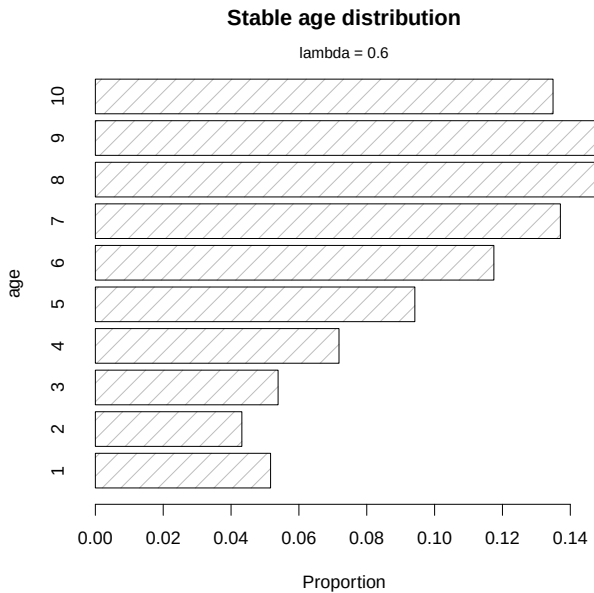
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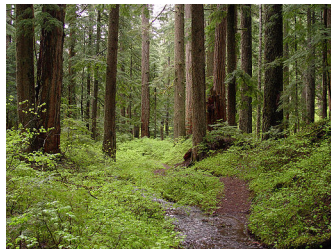
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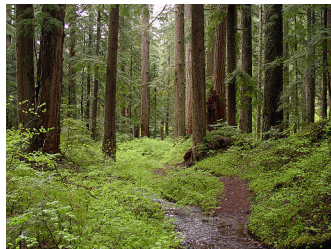
Forest example

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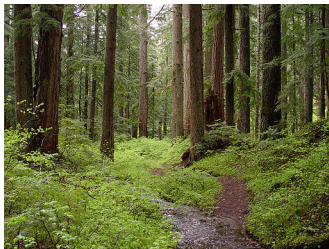
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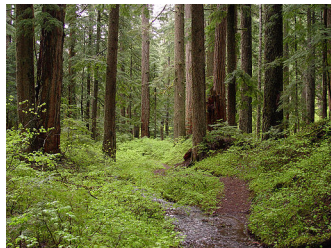
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- ▶ How is it possible that these systems are dominated by large trees?
 - ▶ * Large trees are larger
 - ▶ * Population may be declining
 - ▶ * Trees may spend longer in some size classes than in others
 - ▶ * Life table may not be constant (smaller trees may recruit at certain times and places)

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